

Therefore KE_B is lower than KE_A .

12

C

$$\begin{aligned}
 \text{heat gained by ice [to melt] to reach } 0^\circ\text{C} &= m_{\text{ice}} c_{\text{ice}} (40) [+ m_w L_f] \\
 &= (0.400)(2100)(40) [+ (0.400)(340000)] \\
 &= 33600 [+ 136000] \\
 &= [169600 \text{ J}]
 \end{aligned}$$

$$\begin{aligned}
 \text{heat lost by water [to form ice] to reach } 0^\circ\text{C} &= m_w c_w (30) + m_p c_p (30) [+ m_w L_f] \\
 &= (0.100)(4200)(30) + (0.150)(170)(30) [+ (0.100)(340000)] \\
 &= 12600 + 765 [+ 34000] \\
 &= 13365 [+ 34000] \\
 &= [47365 \text{ J}]
 \end{aligned}$$

At 0°C ,

1. Heat required for ALL ice to melt $\gg$ heat loss when ALL water has formed ice.
2. This means, that water and ice co-exist.